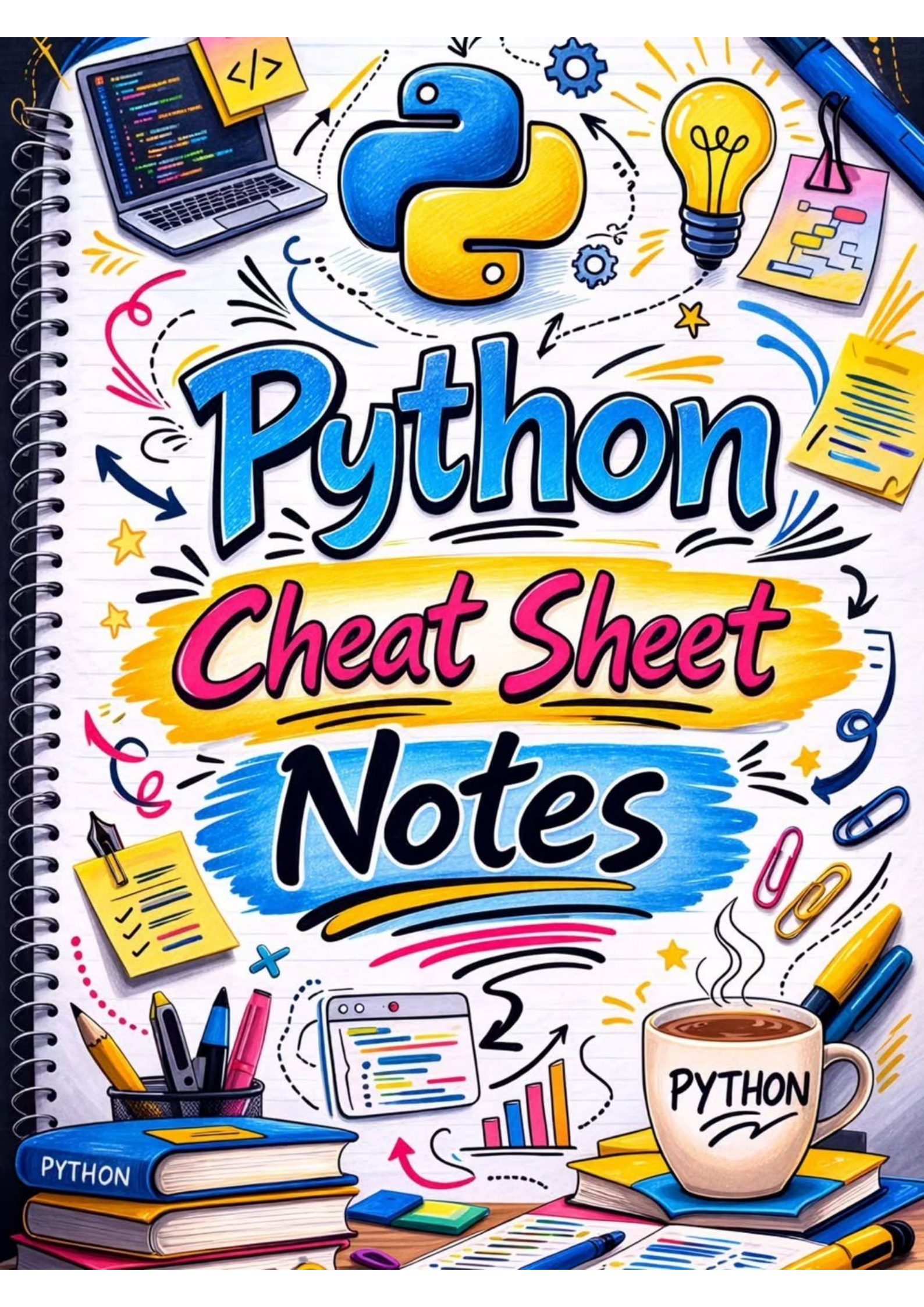




Python

Cheat Sheet

Notes

Introduction to Python

What is Python?

Python is a high-level, interpreted programming language.

Easy to read & write!



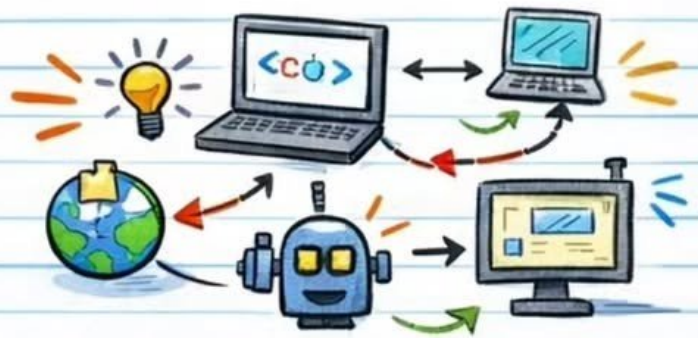
Used in:

Web, Data Science,
AI & More!



Features of Python

- ✓ Easy to Learn
- ✓ Open Source
- ✓ Interpreted (No Compilation!)
- ✓ Huge Libraries
- ✓ Works on All Platforms



Applications of Python

- Web Development
- Data Science
- Machine Learning
- Automation
- Game Development



Basic Python Syntax

```
print("Hello World!")
```

→ Prints output to the screen!



Python Variables and Data Types

Variables

Variables are used to store data values.

Example

`name = "John" age = 22`

Python automatically detects the data type.

Common Data Types

1 Integer

Whole numbers.

→ `x = 10`



2 Float

Decimal numbers.

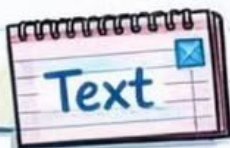
→ `price = 10.5`



3 String

Text values.

→ `name = "Python"`



4 Boolean

True or False values.

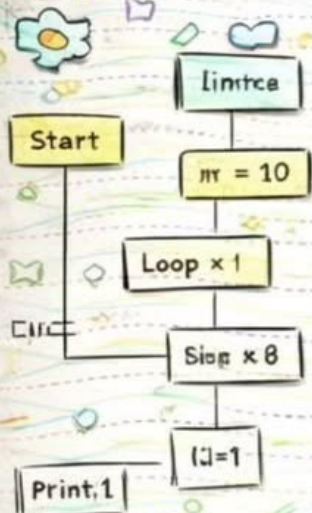
→ `isActive = True`



Check Data Type

`print(type(x))`

→ `class 'int'`



Comment

“python”

to Get this Pdf



- Seop
- × Loop
- ×126
- Print(nx)



Python Operators



Operators are used to perform operations on variables and values.

1. Arithmetic Operators

Used for mathematical operations.

Examples



$a + b$ | $a - b$ | $a * b$ | a / b



2. Comparison Operators

Used to compare values.

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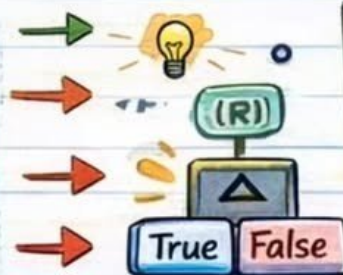
$a == b$ | $a != b$
 $a > b$ | $a < b$



3. Logical Operators

Used to combine conditions.

$a == b$
 $a != b$
 $a > b$
 $a < b$



$x = 10$
 $y = 20$

`print(x < y and y > 10)`

True

Example

$x = 10$
 $y = 20$

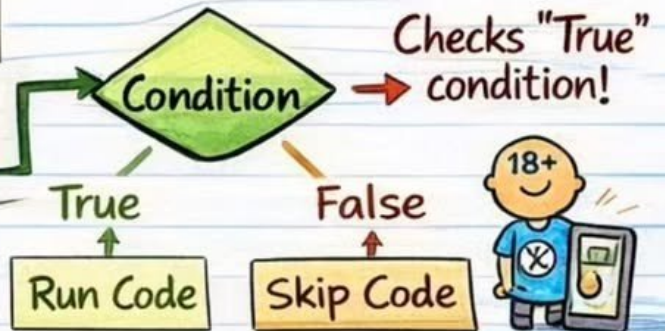
Conditional Statements

Conditional statements are used to make decisions in a program.

if Statement

```
age = 18
```

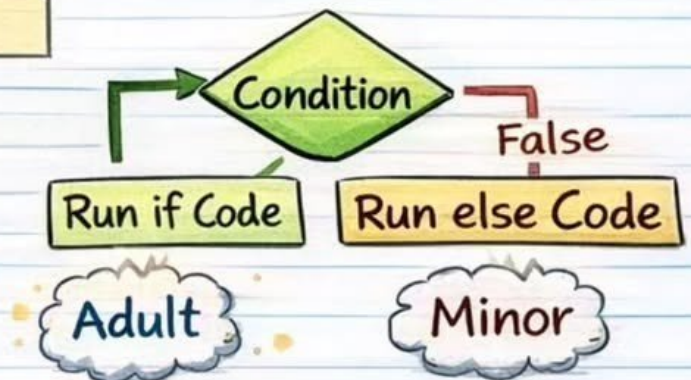
```
if age >= 18:  
    print("Eligible to vote")
```



if-else Statement

```
age = 16
```

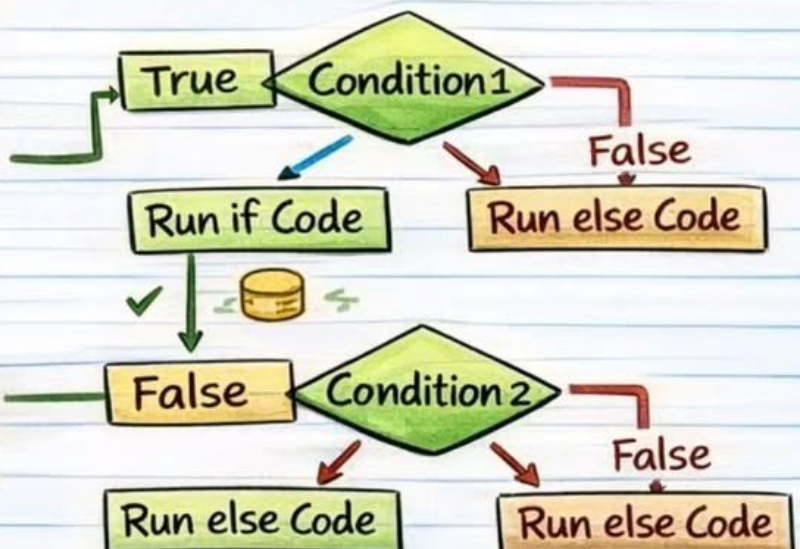
```
if age >= 18:  
    print("Adult")  
else:  
    print("Minor")
```



if-elif-else Statement

```
marks = 75
```

```
if marks >= 90:  
    print("A Grade")  
elif marks >= 60:  
    print("B Grade")  
else:  
    print("C Grade")
```



Python Loops



Loops are used to repeat a block of code multiple times.

1. for Loop

Used to iterate over a sequence.



Example

```
for i in range(5):  
    print(i)
```

Output

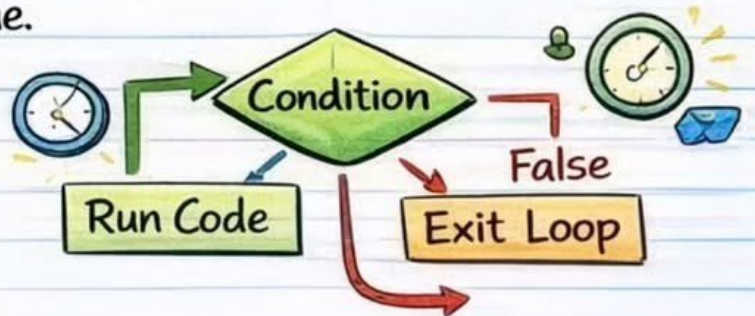
0
1
2
3
4



2. while Loop

Runs while a condition is true.

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```



Loop Control Statements



break → stop loop



skip iteration.



continue → skip iteration



placeholder



pass → placeholder

Python Functions



Functions are reusable blocks of code used to perform specific tasks.

1 Creating a Function

Used to iterate over a sequence.

```
def greet():  
    print("Hello")
```

Function Definition



2 Calling a Function

```
greet()
```

Output → "Hello"

Condition

Run Code

Exit Loop

2 Function with Parameters

```
def add(a, b):  
    return a + b
```

Output → "Hello"

Pass Arguments

Output → 8

Benefits of Functions

- ✓ Code reusability
- ✓ Better organization
- 🔍 Easier debugging



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- 📦 Better organization
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Python Lists, Tuples and Dictionaries



[] ~ []

List: Ordered and mutable collection.

List

Ordered and mutable collection.

Example

```
numbers = [1, 2, 3, 4]
```

0	1	2	3	4
-	0	1	2	3

Output: 1

[] Access element

```
print(numbers(0))
```

Syntax:

```
def function_name():  
    # code block.
```

Tuple

Ordered but immutable collection.

Example

```
{10, 20, 30} [ ]
```

→ Cannot change values.

Dictionary

Stores data as key-value pairs.

```
student = {  
    "name": "John",  
    "age": 20
```



Access value

```
print(student["name"])
```



```
print(add(5, 3))
```

→ Stores data as key-value pairs.



= Code Reusability



= Better Organization

Output: John



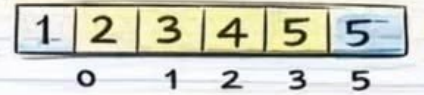
Function

Important Python Libraries

Python provides many powerful libraries.

NumPy

Used for numerical computing and arrays.



Example



NumPy

Used for numerical computing and arrays.



Pandas

Used for data analysis and data manipulation.



Example

matplotlib

Used for data visualization and charts.



Scikit-learn

Used for machine learning algorithms.



TensorFlow

Used for deep learning and AI models.

Flask / Django

Used for building web applications.



Requests

Flask | django



Python is one of the most in-demand programming languages and widely used in modern software development, AI, and data science.

